Box 39.1 Congenital Causes of Urinary Tract Obstruction

Ureteropelvic Junction

Ureteropelvic junction obstruction
Proximal and middle ureter
Ureteral folds
Ureteral valves
Strictures
Benign fibroepithelial polyps
Retrocaval ureter

Distal Ureter

Ureterovesical junction obstruction
Vesicoureteral reflux
Prune-belly syndrome
Ureteroceles

Bladder

Bladder diverticula
Neurologic conditions (e.g., spina bifida)

Urethra

Posterior urethral valves
Urethral diverticula
Anterior urethral valves
Urethral atresia
Labial fusion

Box_39_1

Box 39.2 Acquired Causes of Urinary Tract Obstruction

Intrinsic Processes

Intraluminal

Intrarenal
Uric acid nephropathy
Sulfonamides
Acyclovir
Indinavir
Multiple myeloma
Intraureteral
Nephrolithiasis
Papillary necrosis
Blood clots
Fungus balls

Intramural

Functional
Diseases
Diabetes mellitus
Multiple sclerosis
Cerebrovascular disease
Spinal cord injury
Parkinson disease
Drugs
Anticholinergic agents
Levodopa (α-adrenergic properties)
Anatomic
Ureteral strictures
Schistosomiasis
Tuberculosis
Drugs (e.g., nonsteroidal antiinflammatory agents)
Ureteral instrumentation
Urethral strictures
Benign or malignant tumors of the renal pelvis, ureter, bladder

Extrinsic Processes

Reproductive Tract

Females
Uterus
Pregnancy
Tumor (fibroids, endometrial or cervical cancer)
Endometriosis
Uterine prolapse
Ureteral ligation (surgical)
Ovary
Tuboovarian abscess
Tumor
Cyst
Males
Benign prostatic hyperplasia
Prostate cancer

Malignant Neoplasms

Genitourinary tract
Tumors of kidney, ureter, bladder, urethra
Other sites
Metastatic spread
Direct extension

Gastrointestinal System

Crohn disease
Appendicitis
Diverticulosis
Chronic pancreatitis with pseudocyst formation
Acute pancreatitis

Vascular System

Arterial aneurysms
Abdominal aortic aneurysm
Iliac artery aneurysm
Venous
Ovarian vein thrombophlebitis
Vasculitides
Systemic lupus erythematosus
Polyarteritis nodosa
Henoch-Schönlein purpura

Retroperitoneal Processes

Fibrosis
Idiopathic
Drug induced
Inflammatory
Ascending lymphangitis of the lower extremities
Chronic urinary tract infection
Tuberculosis
Sarcoidosis
Iatrogenic (multiple abdominal surgical procedures)
Enlarged retroperitoneal nodes
Tumor invasion
Tumor mass
Hemorrhage
Urinoma

Biologic Agents

Actinomycosis

Box_39_2

Box 39.3 Effects of Urinary Tract Obstruction on Renal Enzymes and Renal Gene Expression

Changes in Energy and Substrate Metabolism

Decreased oxygen consumption
Decreased substrate uptake
Increased anaerobic glycolysis
Decreased ATP/(ADP + AMP)
Decreased ammonia genesis

Changes in Enzyme Activity

Decreased
Alkaline phosphatase
Na^+/K^+-ATPase
Glucose-6-phosphatase
Succinate dehydrogenase
NADH/NADHP dehydrogenase
Increased
Glucose-6-phosphate dehydrogenase
Phosphogluconate dehydrogenase
Mitogen-activated protein kinases
Matrix metalloproteinase-2/-9
Mast cell protease 1 (chymase)

Changes in Gene Expression

Reduction in glomerular $G_{\alpha s}$ and $G_{\alpha q/11}$ proteins
Reduction in preproepidermal growth factor and Tamm-Horsfall protein
Transient induction of growth factors FOS and MYC
Striking induction of cellular damage *(TRPM2)* genes
Induction of plasminogen activator gene

ADP, Adenosine diphosphate; *AMP,* adenosine monophosphate; *ATP,* adenosine triphosphate; *NADH,* nicotinamide adenine dinucleotide (reduced).

Box_39_3

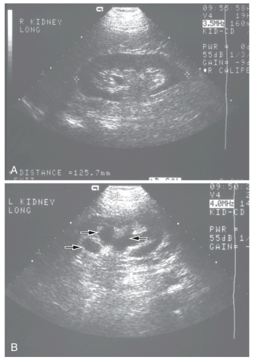

Fig. 39.1 Renal ultrasound. (A) Normal kidney. (B) Hydronephrotic kidney: dilated calices and pelvis *(arrows)*.

Fig_39_1

Fig. 39.2 Intravenous pyelography. Normal right kidney and dilated collecting system on the left. The obstruction was relieved with a stent.

Fig_39_2

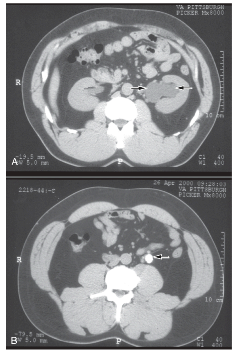

Fig. 39.3 Computed tomography, noncontrast study. (A) Left hydronephrosis: dilated renal pelvis *(arrows)*, with normal kidney on the right. (B) Reason for obstruction: left midureteral stone *(arrow)*.

Fig_39_3

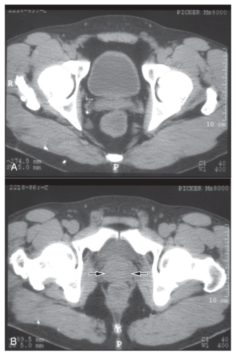

Fig. 39.4 Computed tomography of the pelvis. (A) Large post-voiding residual urine in the bladder. (B) Enlarged prostate *(arrows)*, leading to urinary retention.

Fig_39_4

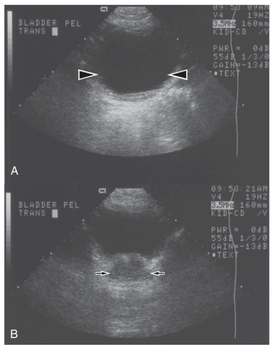

Fig. 39.5 Pelvic ultrasound. (A) Distended bladder *(arrowheads)*. (B) Enlarged prostate *(arrows)*, causing infravesical urinary obstruction.

Fig_39_5

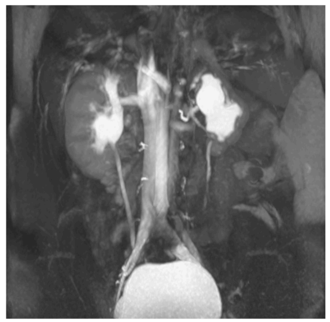

Fig. 39.6 Magnetic resonance (MR) imaging urography of left-sided hydronephrosis with parenchymal thinning. MR urographic image shows large dilatation of the left pelvicalyceal system and narrowing of the left ureteropelvic junction segment.

Fig_39_6

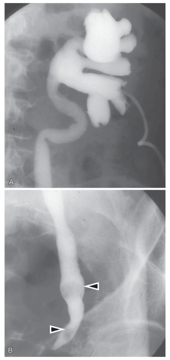

Fig. 39.7 **Antegrade pyelography.** (A) Dilated renal pelvis and calices on left. (B) Stones *(arrowheads)* as filling defects in the distal ureter (not seen on plain film). Intravenous pyelography was unsuccessful owing to the obstructed and malfunctioning kidney.

Fig_39_7

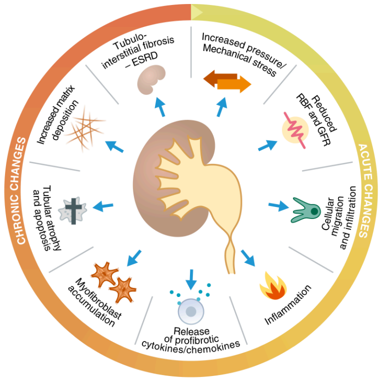

Fig. 39.8 Hallmarks of obstructive nephropathy showing key characteristics of functional and cellular consequences of urinary tract obstruction as a time lapse from acute to more chronic conditions. *ESRD,* End-stage renal disease; *GFR,* glomerular filtration rate; *RBF,* renal blood flow. (Figure modified related to Nørregaard R, Mutsaers HAM, Frøkiær J, Kwon TH. Obstructive nephropathy and molecular pathophysiology of renal interstitial fibrosis. *Physiol Rev.* 2023;103:2827–2872.)

Fig_39_8

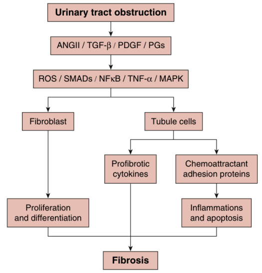

Fig. 39.9 Schematic overview of main signaling pathways involved in urinary obstruction-induced fibrogenesis. Binding of various factors to their receptors activates inflammatory and profibrotic signaling pathways, resulting in the formation and deposition of extracellular matrix (ECM) proteins. Prostaglandins (PGs) can be either profibrotic or antifibrotic depending on which receptor is activated. ANG II, angiotensin II; MAPK, mitogen-activated protein kinase; NF-κB, nuclear factor (NF)-κB; PDGF, platelet-derived growth factor; ROS, reactive oxygen species; TGF-β, transforming growth factor-β; TNF-α, tumor necrosis factor-α.

Fig_39_9

Table 39.1 Glomerular Hemodynamics in Ureteral Obstruction[a]

Stage of obstruction	P_T	R_A	P_{GC}	SNGFR
1-2 hours unilateral	↑↑	↓	↑	=
24 hours unilateral	=	↑↑	↓	↓↓
24 hours bilateral	↑↑	=	=	↓↓
After release: 24 hours unilateral	↓	↑↑	↓↓	↓↓
After release: 24 hours bilateral	=	↑↑	↓	↓↓

[a]See text for discussion and references.
P_{GC}, Hydraulic pressure of Bowman space; *P_T,* proximal tubule hydraulic pressure; *R_A,* afferent arteriole resistance; *SNGFR,* single-nephron glomerular filtration rate; =, unchanged; ↑, increased; ↑↑, markedly increased; ↓, reduced; ↓↓, markedly reduced.

Table_39_1

Table 39.2 Segmental Reabsorption in Superficial and Juxtamedullary Nephrons and in Collecting Ducts in Normal Rats After Release of Bilateral or Unilateral Obstruction

	Normal		After unilateral obstruction		After bilateral obstruction	
Site	Water remaining (%)	Na^+ remaining (%)	Water remaining (%)	Na^+ remaining (%)	Water remaining (%)	Na^+ remaining (%)
S_1	100	100	100	100	100	100
S_2	44	44	26	26	45	45
S_3	26	14	21	12	40	22
S_4	9.4	5	3.2	1.9	25	7
J_1	100	100	100	100	100	100
J_2	12	40	42	52	42	62
CD_1	3.3	2	4.2	3.8	8	6
CD_2	0.4	0.6	2.9	2.5	16.7	12

In obstruction, increased proportions of filtered salt and water are delivered to the loop of Henle in juxtamedullary nephrons (J_1 and J_2 indicating decreased reabsorption). Delivery of sodium and water to the first accessible portion of the inner medullary collecting duct, labeled CD_1, was also increased, and net sodium and water reabsorption along the inner medullary collecting duct (between CD_1 and CD_2) was diminished in both bilateral and unilateral obstruction. In bilateral obstruction, there was net addition or secretion of sodium and water into the lumen of the inner medullary collecting duct, suggesting that in this setting the inner medullary collecting duct secretes sodium and water.
CD_1, Collecting duct at base of papilla, first accessible portion of inner medullary collecting duct; *CD_2,* end of collecting duct as it opens into renal pelvis; *J_{1-2},* values found in juxtamedullary nephrons; *J_1,* Bowman space; *J_2,* tip of loop of Henle; *S_{1-4},* values found in superficial nephrons: *S_1,* Bowman space; *S_2,* end of the proximal convoluted tubule; *S_3,* earliest portion of the distal tubule; *S_4,* end of the distal tubule/beginning of collecting duct.
(Data from Harris KP, Schreiner GF, Klahr S. Effect of leukocyte depletion on the function of the postobstructed kidney in the rat. *Kidney Int*. 1989;36:210–215.)

Table_39_2

Table 39.3 Function of Isolated Perfused Tubules in Obstructive Nephropathy

	J_v SPCT (nL/mm/min)	J_v PST (nL/mm/min)	ΔCl^- MTAL (mEq/L)	J_v CCT (ADH) (nL/mm/min)
Control	0.75 ± 0.08	0.25 ± 0.02	–37 ± 3	0.90 ± 0.08
Unilateral obstruction	0.73 ± 0.11	0.12 ± 0.03	–9 ± 1	0.22 ± 0.04
Bilateral obstruction	0.80 ± 0.08	0.16 ± 0.02	–10 ± 1	0.23 ± 0.04

The J_V in the SPCT was not affected by obstruction, whereas J_V in PST decreased by 52% in unilateral obstruction (0.12 ± 0.03 vs. 0.25 ± 0.02 nL/mm/min) and similarly in response to bilateral obstruction. In mTAL the ability to lower the perfusate chloride ion concentration was reduced by 76% (–9 ± 1 vs. –37 ± 3 mEq/L) and similarly in response to bilateral ureteral obstruction. After relief of unilateral obstruction, the ability of the CCT to respond to ADH was reduced by 76% (0.22 ± 0.04 vs. 0.90 ± 0.08 nL/mm/min) and similarly following relief of bilateral obstruction.
ADH, Antidiuretic hormone; *CCT,* cortical collecting tubule; *J_v,* net fluid reabsorption rate per length of the tubule segment; *PST,* cortical proximal straight tubule; *SPCT,* superficial proximal convoluted tubule; *ΔCl^- MTAL,* change in Cl^- concentration per length of the medullary thick ascending limb.
(Data from Reyes AA, Martin D, Settle S, Klahr S. EDRF role in renal function and blood pressure of normal rats and rats with obstructive uropathy. *Kidney Int*. 1992;41:403–413.)

Table_39_3